Preliminary exercises: Statistical basics

In this session, we will look into examples of basic mathematical tools available to describe random elements. A set of various exercises are proposed. An autonomous study will be made using Matlab to realize the information contained in the different mathematical entities, and their respective limits.

1 A random experiment

1. Simulate a random experiment:

   - Let us consider the suitcases embarked on a plane.
     - First, consider different flights with a random number of suitcases:
       * choose a sample space of the experiment, e.g. [100,150], and a uniform probability distribution. Calculate the probability mass function and its probability cumulative mass function.
       * simulate $n_1 = 10$, and $n_2 = 100$ numerical experiments of this problem,
       * choose a specific value of the outcome: what is the probability to get this outcome with respect to the set of 10 numerical experiments with respect to the chosen probability distribution?
       * repeat the simulation for a normal distribution ($\mu = 125, \sigma = 5$) and a Weibull distribution.
     - Next, consider a set of suitcases with random masses:
       * choose a sample space of the experiment, a probability distribution (represent its probability function and its probability cumulative function),
       * simulate $n_1 = 10$, and $n_2 = 100$ numerical experiments of this problem,
       * choose a specific value of the outcome: what is the probability to get this outcome with respect to the set of 10 numerical experiments with respect to the chosen probability distribution?

2 Moments of probability distribution

Moments allow to exhibit some typical distribution properties of a random variable, which can be discrete or continuous. In practice, even continuous random variables will be in general numerically represented (and so approximated) as discrete random variables.

2.1 For discrete variables

This exercise is proposed for the simplest case of discrete random variables, which have their sample space included in $\mathbb{R}$:

2. Write a Matlab function which computes the empirical mean (`Mean`) for a discrete random variable.

3. Write a Matlab function which computes the empirical variance (`Variance`) and the standard deviation for a discrete random variable.

4. Write a Matlab function which computes the empirical skewness (`Skewness`) for a discrete random variable.

5. Write a Matlab function which computes the empirical kurtosis (`Kurtosis`) for a discrete random variable.

6. Check that the results provided by the previous questions are equal to the ones of the predefined functions in Matlab: `mean`, `var`, `skewness` and `kurtosis`.

7. Provide some sets of data:
   - with same mean, same skewness but different variances,
   - with same mean, different variance, and different skewness,
   - with positive skewness,
   - with negative skewness,
   - with positive kurtosis.
8. Which of these moments exhibit linear properties?

2.2 For some classical continuous distributions

9. For a uniform distribution law, establish the analytical expressions of the expected value, the variance, the skewness and the kurtosis.

3 A Gaussian continuous distribution

10. Investigate a Gaussian continuous distribution:
    - Plot Gaussian probability density function and Gaussian cumulative distribution function for different mean values,
    - Plot Gaussian probability density function and Gaussian cumulative distribution function for different values of the standard deviation of the random variable,
    - Which are the skewness and the kurtosis of these different distributions?
    - From the probability density function or from the cumulative distribution function, determine the probability that x belongs to $[a; b]$.

4 Mean, median and mode

11. Compute the mean, the median and the mode for the discrete set of data: $\{2; 2; 2; 15; 36; 10\}$.
12. What are the mean, the median and the mode for a Gaussian distribution?